

Opta Physical I/O Map

Duplex Lift Station + EZ Meter (DUPLEXLS)

ST program: logic/38_duplex_lift_station_ezmeter.st
MooreVIEW Opta firmware: 2.3.77
Generated: 2026-08-02

Hardware

- Arduino Opta (base I1–I8 digital, I1–I6 analog, R1–R4 relay)
- Sequent D1608E digital expansion — slot 1 (X1_I1...X1_I16, X1_R1...X1_R8)
- Sequent A0602 analog expansion — slot 2 optional (X2_AI1...X2_AI8, X2_PWM1...X2_PWM4)

Wiring notes

- Expansion DIs (+24 V active): input ON when float wet, run aux made, or fault contact active. Unwired inputs read OFF.
- Run feedback and fault inputs: +24 V active (input ON when aux made or fault contact active).
- Analog I1–I6: 0–1 V CT transmitters, linear 0–50 A (1.00 V = 50 A). Engineering tag = raw × 0.48828125.
- Relay outputs R1/R2 drive pump contactors; R4 drives station alarm horn/strobe (LVL_HIGH, MECH_PQ_ALM, motor fault DIs).
- EZ Meter DDS-RGB on Opta RS485 (A/B, 9600 8N1, slave ID 1). Flash with MV_FIELDBUS=1 MV_EZMETER=1.
- Float DIs: HIGH (X1_I1) · LEAD (X1_I2) · LAG (X1_I3) · OFF (X1_I4).
- PHASE_FAULT (X1_I7) clears auto pump demand in ST — same effect as dry off float while upper floats are on.

Opta base — relay outputs

Terminal / tag	Point	Type	Driver channel	Logic tag	Description
R1	Relay out 1	BOOL	R1	MOTOR1_RUN	Pump 1 contactor
R2	Relay out 2	BOOL	R2	MOTOR2_RUN	Pump 2 contactor
R3	Relay out 3	BOOL	R3	—	Spare (triplex: pump 3 contactor)
R4	Relay out 4	BOOL	R4	R4	Station alarm (high level / EZ Meter PQ / motor fault)

Opta base — digital inputs (spare)

Terminal / tag	Point	Type	Driver channel	Logic tag	Description
I7	Digital in 7	BOOL	I7	—	Spare (I1–I6 used as analog CT inputs)
I8	Digital in 8	BOOL	I8	—	Spare

Opta base — analog inputs (CT 0–50 A)

Terminal / tag	Point	Type	Driver channel	Logic tag	Description
AI1 / I1	Analog in 1	REAL	I1_RAW	AI1	Motor 1 phase A amps
AI2 / I2	Analog in 2	REAL	I2_RAW	AI2	Motor 1 phase B amps
AI3 / I3	Analog in 3	REAL	I3_RAW	AI3	Motor 1 phase C amps
AI4 / I4	Analog in 4	REAL	I4_RAW	AI4	Motor 2 phase A amps
AI5 / I5	Analog in 5	REAL	I5_RAW	AI5	Motor 2 phase B amps
AI6 / I6	Analog in 6	REAL	I6_RAW	AI6	Motor 2 phase C amps
I7_RAW	Analog raw 7	INT	I7_RAW	—	Spare raw (if I7 configured analog)

I8_RAW	Analog raw 8	INT	I8_RAW	—	Spare raw (if I8 configured analog)
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Expansion slot 1 — Sequent D1608E digital inputs

Terminal / tag	Point	Type	Driver channel	Logic tag	Description
X1_I1	DI 1	BOOL	X1_I1	LVL_HIGH	High level float
X1_I2	DI 2	BOOL	X1_I2	LVL_LEAD	Lead float
X1_I3	DI 3	BOOL	X1_I3	LVL_LAG	Lag float
X1_I4	DI 4	BOOL	X1_I4	LVL_OFF	Off float
X1_I5	DI 5	BOOL	X1_I5	P1_RUN_FB	Pump 1 run feedback (fail-to-run)
X1_I6	DI 6	BOOL	X1_I6	P2_RUN_FB	Pump 2 run feedback (fail-to-run)
X1_I7	DI 7	BOOL	X1_I7	PHASE_FAULT	Phase fault
X1_I8	DI 8	BOOL	X1_I8	GEN_RUN	Generator running
X1_I9	DI 9	BOOL	X1_I9	GEN_FUEL_FAULT	Generator fuel fault
X1_I10	DI 10	BOOL	X1_I10	GEN_FAULT	Generator fault
X1_I11	DI 11	BOOL	X1_I11	MOTOR1_FAULT	Pump 1 motor fault
X1_I12	DI 12	BOOL	X1_I12	MOTOR2_FAULT	Pump 2 motor fault
X1_I13	DI 13	BOOL	X1_I13	—	Spare
X1_I14	DI 14	BOOL	X1_I14	—	Spare
X1_I15	DI 15	BOOL	X1_I15	—	Spare
X1_I16	DI 16	BOOL	X1_I16	—	Spare

Expansion slot 1 — Sequent D1608E relay outputs (spare)

Terminal / tag	Point	Type	Driver channel	Logic tag	Description
X1_R1	Relay out 1	BOOL	X1_R1	—	Spare (pump contactors on base R1/R2)
X1_R2	Relay out 2	BOOL	X1_R2	—	Spare
X1_R3	Relay out 3	BOOL	X1_R3	—	Spare
X1_R4	Relay out 4	BOOL	X1_R4	—	Spare
X1_R5	Relay out 5	BOOL	X1_R5	—	Spare
X1_R6	Relay out 6	BOOL	X1_R6	—	Spare
X1_R7	Relay out 7	BOOL	X1_R7	—	Spare
X1_R8	Relay out 8	BOOL	X1_R8	—	Spare

Expansion slot 2 — Sequent A0602 analog (optional RTD)

Terminal / tag	Point	Type	Driver channel	Logic tag	Description
X2_AI1	Analog 1	REAL	X2_AI1	MOTOR1_TEMP	Motor 1 winding temp (cal in Tags)
X2_AI2	Analog 2	REAL	X2_AI2	MOTOR2_TEMP	Motor 2 winding temp (cal in Tags)
X2_AI3	Analog 3	REAL	X2_AI3	—	Spare
X2_AI4	Analog 4	REAL	X2_AI4	—	Spare
X2_AI5	Analog 5	REAL	X2_AI5	—	Spare
X2_AI6	Analog 6	REAL	X2_AI6	—	Spare
X2_AI7	Analog 7	REAL	X2_AI7	—	Spare
X2_AI8	Analog 8	REAL	X2_AI8	—	Spare

Expansion slot 2 — Sequent A0602 PWM outputs (spare)

Terminal / tag	Point	Type	Driver channel	Logic tag	Description
X2_PWM1	PWM 1	REAL	X2_PWM1	—	Spare
X2_PWM2	PWM 2	REAL	X2_PWM2	—	Spare
X2_PWM3	PWM 3	REAL	X2_PWM3	—	Spare
X2_PWM4	PWM 4	REAL	X2_PWM4	—	Spare

Related documents

Live values: MooreVIEW Tags !' Live I/O, or Opta <http://<opta-ip>/io-map> when on Ethernet.

ST logic reference: st/logic/36_duplex_lift_station.st (PHYSICAL I/O MAP header).